Muhammad Zahid

Data Engineer - Data, Platform and AI

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Eligible to work in the EU

SUMMARY

Data engineer with 4+ years building and owning production data pipelines end to end in Python and SQL - ingestion, ETL/ELT transformation, data quality testing, monitoring and reliability - across AWS and Azure. Production use of Kafka, dbt, Airflow, PySpark, Databricks, ClickHouse and Redshift, with a track record of concrete performance wins: streaming latency cut from 45 minutes to under 90 seconds, 1000x Kafka consumer throughput and 70% faster analytics queries.

PROFESSIONAL EXPERIENCE

Data and Platform Engineer, Salla **09/2025 – Present | Remote, Finland**

- Re-architected a legacy batch pipeline into a streaming system on Kafka + Timeplus, cutting end-to-end data latency from 45 minutes to under 90 seconds for 1.2B+ daily events
- Migrated core analytics workloads from a self-managed Postgres cluster to ClickHouse on AWS, reducing query times by 70% for dashboards used by 200+ internal stakeholders
- Diagnosed and fixed a production Kafka consumer bottleneck through structured root-cause analysis, increasing throughput 1000x (~25 to ~25,000 msg/s) by rewriting microservice logic, adding partitions and enabling KEDA-based horizontal autoscaling
- Built and owned 12 ELT pipelines (Mage AI + dbt) integrating ERP, transactional and event data into a ClickHouse warehouse, replacing a manual financial reconciliation process; built staging models, dbt tests and macros for repeatable data-quality checks
- Owned ClickHouse (OLAP) administration end to end - schema design (ORDER BY, partitioning, skip indexes), query and performance tuning and 2-replica high availability across terabytes of data - lifting average query performance ~30%
- Wrote dbt tests and Python pytest suites wired into GitHub Actions CI/CD so every deploy was validated against data-quality and code checks; instrumented services with Graylog logging and Slack alerting for monitoring and rapid incident response
- Built, containerized (Docker) and deployed event-driven Python microservices consuming Kafka streams with Redis, authoring Helm charts and managing Kubernetes deployments with KEDA autoscaling - sustaining ~98% uptime under variable load
- Served as onboarding buddy for a newly joined colleague, mentoring them through the data platform, codebase and team practices

Data Engineer, Addo AI **08/2023 – 09/2025 | Remote, Pakistan**

- Led architecture of a government analytics platform processing several terabytes of multi-sectoral data daily across 7+ source systems (~200 Airflow DAGs), building an on-prem bare-metal lakehouse on Delta Lake/MinIO, Trino, ClickHouse and MySQL with Apache Superset to serve 7+ departmental teams
- Built ELT pipelines on Azure Databricks for a global consumer-health client, ingesting Excel, REST API and email data with incremental loading into ADLS Gen2, and set up Unity Catalog catalogs and schemas to organise the lakehouse tables
- Built a generic Python ETL framework for ingestion from many inconsistent third-party sources - REST APIs, MySQL, SQL Server, web scraping and PDFs - cutting deployment time 60% and end-to-end processing time from 10 to 5 hours, meeting a 7-hour SLA
- Implemented real-time streaming (Kafka + PySpark) and dimensional models (fact/dimension tables, SCD Type 2/4, geographic entity normalization) across 7+ source systems
- Led ClickHouse adoption as the OLAP engine, delivering a 40% query performance gain on multi-terabyte workloads
- Migrated a legacy Oracle 19c system to AWS, building the ETL in Glue jobs and landing curated data in Redshift as the target warehouse
- Led a team of 3 engineers delivering BI dashboards and Python-based ingestion automation (API clients, DAG orchestration), enforcing code review as a standing practice, using Git and Azure DevOps CI/CD pipelines

Software Engineer, Bitnine Global **08/2022 – 08/2023 | Seoul, South Korea**

- Read, debugged and extended an existing production C/C++ codebase: developed core features for Apache AGE, an open-source graph database extension for PostgreSQL, and contributed to PG-Pool II
- Served as main developer for the AgensSQL Migration Tool, taking features from design through implementation and testing

PROJECTS

PITB OSSP Data Platform (Addo AI)

Stack: Airflow, Delta Lake, Trino, ClickHouse, Superset, MinIO

- Designed and implemented an on-prem government analytics platform under data-sovereignty rules: many sectors, terabytes a day, no public cloud warehouse
- Modeled a curated star schema with Trino on Delta Lake/MinIO, with ClickHouse (OLAP) serving thin BI layers, decoupling data modeling from reporting performance

Data Modelling Agent (Personal project)

Stack: Claude Code, Python, Oracle 11g, Redshift, S3, Tableau

- Generates logical data models (LDMs) with auto-healing using Claude, with guardrailed agent personas and observability logging every step and decision

SKILLS

Languages: Python, SQL, JavaScript, Bash, C/C++

Streaming & Orchestration: Kafka, Spark, PySpark, Airflow, Mage AI, Timeplus

Databases & Warehousing: ClickHouse (OLAP), Redshift, PostgreSQL, Delta Lake, Trino, Iceberg, MySQL, SQL Server, Oracle, MongoDB, Redis

AWS: S3, Glue, Lambda, SQS, EKS, Redshift

Azure: Azure Databricks, Azure Data Factory, ADLS Gen2, Azure SQL Database, Azure Functions, Azure DevOps

Data Modeling & Quality: dbt (models, tests, macros), Star/Snowflake schemas (dimensional modeling), SCD Type 2/4, schema evolution, data contracts, data marts

Testing & CI/CD: pytest, dbt tests, GitHub Actions, GitLab CI/CD, Azure DevOps, code review

Infrastructure: Docker, Kubernetes, Helm, KEDA, Linux, GitOps, Terraform (working)

Observability: Grafana, Graylog, Slack alerting, query tuning (EXPLAIN)

Backend & APIs: REST APIs, FastAPI, Flask, microservices, event-driven architecture

BI & AI: Power BI, Superset, Tableau, LLM integration, Claude Code, AI agents

EDUCATION

MSc Computer Science, Åbo Akademi University **08/2025 – 05/2027 (expected) | Turku, Finland**

BSc Computer Science, National University of Computer and Emerging Sciences (FAST NUCES) **08/2019 – 06/2023 | Islamabad, Pakistan**

LANGUAGES

English (professional), Urdu (native)